Nick Rovito

CONTACT INFORMATION

Department of Mechanical Engineering
ECME 217, University of Colorado, Boulder
Boulder, CO 80309-0427

Phone: (972) 674-7755
E-mail: nick.rovito@colorado.edu

RESEARCH INTERESTS

- **Biomedical:** Computational modeling of biofluids and biomechanics; hemodynamic modeling of the heart-to-brain pathway; stroke and thrombosis modeling; multiscale biochemical reaction modeling in drug transport. Cerebrovascular flow; Biomechanics of cardiovascular diseases.
- **Computational:** Multiphysics and multiscale modeling; fluid-structure interactions; multidomain methods; reduced-order methods; Finite Element Method; Fictitious Domain Method; Discrete Element Method; high-performance computing; multiphysics software development; AI use in engineering pedagogy.

EDUCATION

University of Colorado, Boulder, CO

- *M.S., Mechanical Engineering: December, 2024*

University of Arkansas, Fayetteville, AR

- *B.S., Mechanical Engineering: May, 2023*
- *B.A., Spanish Language: May, 2023*
- *Minor, Mathematics: May, 2023*

AWARDS & HONORS

1. **Overall Best Poster: Runner-Up;** *MechBio Symposium, Boulder, Colorado; May, 2026*
2. **Singh Graduate Fellowship**; *Fall 2025*
3. **National Science Foundation Graduate Research Fellowship**; *2025 – 2028*
4. **1st Place in the ASME International Mechanical Engineering Congress & Exposition (IMECE) Fluids Engineering Division Young Engineering Paper Competition;** *ASME IMECE 2024, Portland, Oregon; November, 2024.*
5. **Outstanding Teaching Assistant Award**; *Awarded to teaching assistants recognized by both faculty members and students for providing exemplary service in teaching support*; *Spring 2024.*
6. **Chair's Graduate Fellowship**; *Fall 2023*
7. **Graduated with High Distinction**; *May, 2023*

JOURNAL ARTICLES

1. **Rovito, N.**, Holmes, J., Teeraratkul, C., and Mukherjee, D. (2026). The FLATiron toolkit: A versatile multiphysics platform for flow and transport phenomena using the finite element method. *SoftwareX*. 35:102968.
2. Laroche, A., **Rovito, N.**, Liu, A., Allaeys, I., Heitsch, L., Portier, I., Adeoye, O., Pizzella, S.R., Giles, J.A., Di Paola, J., Lee, J.-M., Bark, D., Campbell, R.A., Mukherjee, D., Boilard, É., and Denorme, F. (2026). Mechanisms of von Willebrand factor activation driving no reflow in ischemic stroke. *Proceedings of the National Academy of Sciences*. 123(30):e2610397123.
3. Venkatesh, S., Teeraratkul, C., **Rovito, N.**, Mukherjee, D., and Lynch, M.E. (2025). High-fidelity computational fluid dynamics modeling to simulate perfusion through a bone-mimicking scaffold. *Computers in Biology and Medicine*. 186:109637.

Peer-reviewed Proceedings

1. **Rovito, N.**, Malloy, J.S., Denorme, F., and Mukherjee, D. Quantitative Analysis of Elongational Flow Exposure for Assessment of von Willebrand Factor Unfolding in Mechanical Circulatory Support. *In: Proceedings of the 9th International Conference on Computational and Mathematical Biomedical Engineering. Kobe, Japan. June 2026.*
2. **Rovito, N.**, and Mukherjee, D. In Silico Analysis of Flow-Mediated Drug Transport for Thrombolytic Therapy in Acute Ischemic Stroke. *In: Proceedings of the ASME International Mechanical Engineering Congress and Exposition (IMECE). Portland, Oregon. November 2024. Volume 8: Fluids Engineering.*
3. **Rovito, N.**, and Walters, D. K. CFD-Based Optimization of the Kinematic Cycle of an Oscillating Foil Energy Harvesting Device. *In: Proceedings of the ASME International Mechanical Engineering Congress and Exposition (IMECE). New Orleans, Louisiana. October–November 2023. Volume 9: Fluids Engineering.*

Abstracts & Presentations

1. **Rovito, N.**, Mukherjee, D. Cerebrovascular Network Anastomosis and Its Role in Fibrinolytic Drug Transport for Stroke. *CU-CSU CCTSI Summit Poster Session, Aurora, Colorado. August, 2026.*
2. **Rovito, N.**, Holmes, J., and Mukherjee, D. The FLATiron Toolkit: A Versatile Open-source Modeling Framework For Flow And Transport Phenomena. *17th World Congress on Computational Mechanics, Munich, Germany. July 2026.*
3. **Rovito, N.**, Pinchuk, V., Denorme, F., Mukherjee, D. Collateral Vessel Geometry and Elongational Hemodynamics: A Computational Study of VWF Activation in Post-Ischemia Reperfusion. *MechBio Symposium 2026, Boulder, Colorado. May, 2026.*
4. Holmes, J., **Rovito, N.**, Mukherjee, D. Investigating Clot-Flow Interactions Using a Fictitious Domain Based Particle-Continuum Method. *MechBio Symposium 2026, Boulder, Colorado. May, 2026.*
5. **Rovito, N.**, Mukherjee, D. Cerebrovascular Network And Flow Effects In Thrombolytic Therapy For Stroke. *USACM Student Chapter Virtual Poster Competition. April, 2026.*
6. Holmes, J., **Rovito, N.**, Mukherjee, D. USACM - A Particle-Continuum Method for Investigating Clot-Flow Interactions. *USACM Student Chapter Virtual Poster Competition. April, 2026.*
7. **Rovito, N.**, Holmes, J., Mukherjee, D. FLATiron: A Hierarchical, Modular, Finite Element Library for Flow Physics and Transport Phenomena. *10th Annual Rocky Mountain Fluid Mechanics Research Symposium, Boulder, Colorado. August, 2025.*
8. **Rovito, N.**, Mukherjee, D. A Computational Multiphysics Framework for Patient-Specific Thrombolysis Modeling. *18th U.S. National Congress on Computational Mechanics (USNCCM), Chicago, Illinois. July, 2025.*
9. Laroche, A., **Rovito, N.**, Liu, A., Allaeys, I., Campbell, R., Adeoye, O., Mukherjee, D., Boilard, E., Denorme, F. Mechanistic Insights into Detrimental VWF Activation during Stroke. *Oral communication presented at the International Society on Thrombosis and Haemostasis (ISTH) Congress, Washington D.C. June, 2025.*
10. **Rovito, N.**, Mukherjee, D. An In Silico Analysis of Flow-Mediated Drug Transport For Thrombolytic Therapies in Acute Ischemic Stroke. *ASME International Mechanical Engineering Congress & Exposition (IMECE), Portland, Oregon. November, 2024.*
11. **Rovito, N.**, Mukherjee, D. An In Silico Model of Flow-Mediated Fibrinolysis in Acute Ischemic Stroke. *9th Annual Rocky Mountain Fluid Mechanics Research Symposium, Boulder, Colorado. August, 2024.*
12. **Rovito, N.**, Walters D.K. CFD-Based Optimization of the Kinematic Cycle of an Oscillating Foil Energy Harvesting Device. *ASME International Mechanical Engineering Congress & Exposition (IMECE), New Orleans, Louisiana. October, 2023.*

Research Experience

- **Computational fluid dynamics and coupled multiphysics modeling for thrombolysis**
  - Modeled flow, thrombolytic drug transport, and biochemical lysis reactions in the cerebrovascular system using patient-specific cerebrovascular geometries from medical image segmentation.
  - Developed reduced- and hybrid-order models of flow and transport, as well as wall compliance in arterial tissue.
  - Developed the modular, extensible, and scalable multiphysics toolkit, *the FLATiron toolkit*, based on FEniCS/DOLFINx, to aid in computational biofluids research and released it as open source.
- **Computational fluid dynamics modeling for hydroelectric energy systems**
  - Designed mathematical models and applied computational fluid dynamics to optimize 8-degree-of-freedom oscillating foil energy-harvesting systems.

Teaching & Academic Experience

- **Teaching Assistant:** University of Colorado, Boulder
  - `MCEN 4231/5231`: Computational Fluid Dynamics, Spring 2025
  - `MCEN 3021`: Introduction to Fluid Mechanics, Spring 2024
  - `MCEN 3021`: Introduction to Fluid Mechanics, Fall 2023
- **STEM Tutor:** Do College Better, Fayetteville AR (Jan 2022 - Aug 2023)
  - `MATH 1213`: Plane Trigonometry
  - `MATH 1284`: Precalculus Mathematics
  - `MATH 2043`: Survey of Calculus
  - `MATH 2554`: Calculus I
  - `MATH 2564`: Calculus II
  - `MATH 2574`: Calculus III
  - `MATH 2584`: Elementary Differential Equations
  - `MEEG 2023`: Dynamics
  - `PHYS 2054`: University Physics I
- **Teaching Assistant:** University of Arkansas, Fayetteville
  - `MATH 3083`: Linear Algebra, Fall 2021

Professional Service

- **American Society of Mechanical Engineers *(ASME)*:** Aug 2024 - Present.

Service & Outreach

- **Graduate AI Advising Committee**
  - Member; 2026 - Present
- **CU Triple E Fair**
  - Student volunteer, Fall 2025
- **MechBio Symposium**
  - Student volunteer, August 2024

Software

- **FLATiron toolkit (FLAT-FE: FLow and Transport using Finite Elements)**
  - **Role:** Lead Developer, Researcher
  - **Link:** https://github.com/flowlabcu/FLATiron
- **vSCAD (Vascular Systems CAD)**
  - **Role:** Lead Developer, Researcher
  - **Link:** https://github.com/flowlabcu/vSCAD

Professional Experience

- **Research Assistant:** University of Colorado, Boulder; Aug 2023 - Present
  - Develop computational multiphysics simulations and modeling workflows using the Finite Element Method for cerebrovascular and biological flows. Applications include Acute Ischemic Stroke, Thrombolysis, Thrombectomy, and post-treatment care. Develop open-source toolkits for multiphysics FEM modeling and complex network geometry creation.
- **Research Assistant:** University of Arkansas, Fayetteville; Aug 2022 - Aug 2023
  - Developed computational tools for optimization of n-degree kinematic parameters for renewable oscillating foil energy harvesting (OFEH) systems.
- **Manufacturing Engineering Internship:** EPIROC, Garland, TX; June 2021 - Aug 2021
  - Improved cab assembly and production, electrical documentation, and engineering drawings using tools such as Creo Illustrate, Visual Factory, and the principals of lean engineering.

References

Prof. Debanjan Mukherjee ***Email:*** debanjan@colorado.edu
Assistant Professor, Paul M Rady Department of Mechanical Engineering
University of Colorado Boulder

Prof. James H. Leylek ***Email:*** jleylek@uark.edu
Professor, Department of Mechanical Engineering
University of Arkansas

Prof. Keith Walters ***Email:*** keithw@uark.edu
Professor, Department of Mechanical Engineering
University of Arkansas